Review: Exponents and radicals

1 As the base is 4 and the power is 5, you multiply 4 by itself 5 times. So $4^5 = 4 \times 4 \times 4 \times 4 \times 4$

2 Check the power. If the power is odd then the answer will be negative because a negative multiplied by itself an odd number of times will give you a negative. So in this case, for $(-7)^3$, the answer is negative.

3

a No

b No

c Yes

d Yes

4 **B**

5

a 3

b 1

c 64

d 16

e 243

f -64

g -648

h -216

i 45

j 40

k 36

l -64

m 81

n -81

o 16

p 256

q -225

r -64

s -324

t 17

u -5

v 3

w 37

x -63

y 91

z -158

6 $39^2\ \text{m}^2$

7

a True

b False

c False

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8

a 2 b 4 c 8 d 10

e 12 f 2 g 10 h 8

i 8 j 4 k 10 l 13

m 15 n -2 o -9 p -11

q -14

Additional questions

9

a 5

b 7

c -6

d -7

10

a 2

b 4

c -3

d -5

11

a 2 b 4 c 5 d 3

e -1 f -3 g -2 h -4

12 3^3

13

a 31 b 49 c 80 d 64

e 125 f 3 g 4 h 72

i $\frac{3}{4}$ j $\frac{8}{9}$ k 1 l 1

m $\frac{4}{5}$ n 1 o $\frac{9}{5}$ p 9

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14

a

i Ralph distributed the parentheses before taking the square.

ii $2\left(3+2\right)^2 = 2\left(5\right)^2 = 2 * 25 = 50$

b

i Ida moved the (9) into the square root without squaring it first.

ii $\sqrt{9}\left(8 \div 4 - 3 \times 4 + 4^2 + 3\right) = \sqrt{9}\left(2 - 12 + 16 + 3\right) = \sqrt{9}\left(9\right) = 3\left(9\right) = 27$

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